Chapter 1

Refraction

1.1 Refraction

The refraction of light is a phenomenon where light changes its direction when it passes from one medium to another. The change in direction is due to a change in the speed of light as it moves from one medium to another. Do not confuse this with rarefaction, which sounds similar and is a similar wave concept for decompression!

1.2 Snell's Law

This phenomenon is explained by Snell's law, which states:

$$n_1 \cdot \sin(\theta_1) = n_2 \cdot \sin(\theta_2) \tag{1.1}$$

where:

- n_1 and n_2 are the indices of refraction for the first and second media, respectively. The index of refraction is the ratio of, c the speed of light in a vacuum to, v the speed of light in the medium, $n = \frac{c}{v}$. It is a dimensionless quantity. $n \geq 1$ for all materials, such that $n_{\text{vacuum}} = 1$ and $n_{\text{air}} \approx 1$. See a list of all mediums here: https://en.wikipedia.org/wiki/List_of_refractive_indices
- θ_1 and θ_2 are the angles of incidence and refraction, respectively. These angles are measured from the normal (perpendicular line) to the surface at the point where light hits the boundary.

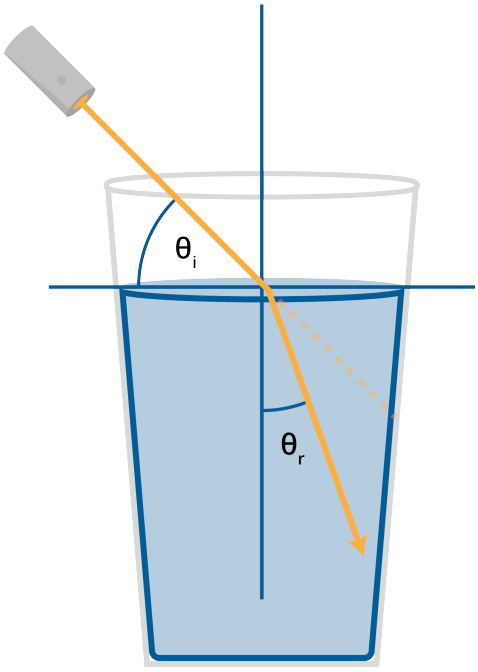


Figure 1.1: Refraction occurs when light changes the medium it is in.

The angle of incidence (θ_1) is the angle between the incident ray and the normal to the interface at the point of incidence. Similarly, the angle of refraction (θ_2) is the angle between the refracted ray and the normal.

When light travels from a medium with a lower refractive index to a medium with a higher refractive index, it bends towards the normal. Conversely, when light travels from a medium with a higher refractive index to one with a lower refractive index, it bends away from the normal.

1.3 Diffraction

This is a draft chapter from the Kontinua Project. Please see our website (https://kontinua.org/) for more details.

Appendix A

Answers to Exercises